



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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0625/62

May/June 2020

1 hour

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

DC (LK/SW) 195016/2
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[Turn over

- 1 A student investigates the period of a pendulum. Fig. 1.1 and Fig. 1.2 show the apparatus she uses.

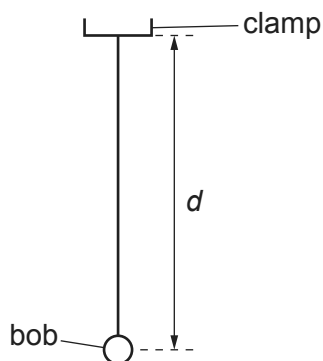


Fig. 1.1

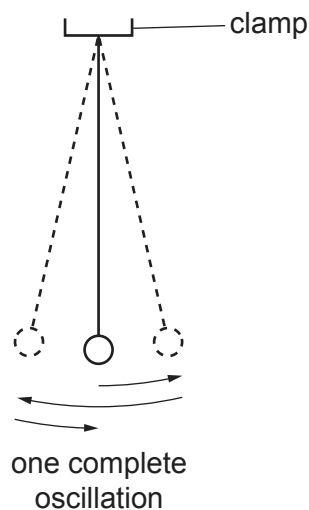


Fig. 1.2

- (a) Explain briefly, with the help of a diagram, how you would use a metre rule and set square to measure the length d of a pendulum as accurately as possible.

Diagram:

.....

.....

..... [3]

- (b) The student adjusts the pendulum so that $d = 50.0$ cm. She displaces the bob slightly and releases it so that it swings. Fig. 1.2 shows one complete oscillation of the pendulum. She measures the time t_1 for 20 complete oscillations.

- (i) Record the time t_1 shown in Fig. 1.3.



Fig. 1.3

$t_1 =$ [1]

- (ii) Calculate the period T_1 of the pendulum. The period is the time for one complete oscillation.

$$T_1 = \dots\dots\dots [1]$$

- (c) The student adjusts the pendulum until the distance d is 100.0 cm.

She repeats the procedure and records the time t_2 for 20 oscillations and the period T_2 .

$$t_2 = \dots\dots\dots 39.80 \text{ s}$$

$$T_2 = \dots\dots\dots 1.99 \text{ s}$$

She measures the mass m_A of the pendulum bob. The reading on the balance is shown in Fig. 1.4.

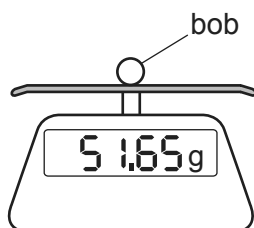


Fig. 1.4

Record mass m_A of the pendulum bob to the nearest gram.

$$m_A = \dots\dots\dots \text{ g } [1]$$

The student repeats the procedure using a pendulum bob of mass m_B .

$$m_B = \dots\dots\dots 109 \text{ g}$$

She obtains these results:

$$\text{distance } d = \dots\dots\dots 50.0 \text{ cm}$$

$$\text{period } T_3 = \dots\dots\dots 1.39 \text{ s}$$

$$\text{distance } d = \dots\dots\dots 100.0 \text{ cm}$$

$$\text{period } T_4 = \dots\dots\dots 2.02 \text{ s}$$

- (d) (i) Using the results T_1 , T_2 , T_3 and T_4 , for the period of each of the pendulums, tick (✓) the response that matches your results within the limits of experimental accuracy.

☐ the period T is affected by d only

☐ the period T is affected by both d and m

☐ the period T is affected by m only

☐ the period T is not affected by d or m

[1]

- (ii) Justify your answer to (d)(i) by reference to the results.

.....

..... [1]

- (e) The student now investigates the effect of the size of the oscillations on the period of the pendulum.

- (i) Suggest briefly how you would measure the size of an oscillation. You may draw a diagram.

.....

..... [2]

- (ii) State **one** variable that you would keep constant during this part of the investigation.

..... [1]

[Total: 11]

- 2 A student determines the resistance of a resistance wire.

Fig. 2.1 shows the circuit he uses.

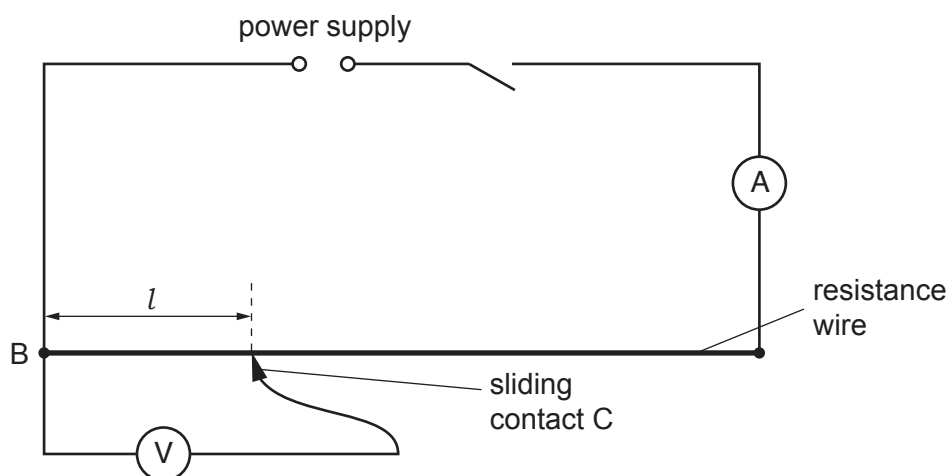


Fig. 2.1

(a)

- The student places the sliding contact C on the resistance wire at a distance $l = 10.0$ cm from B.
- Record, in the first row of Table 2.1, the potential difference V across the length $l = 10.0$ cm of resistance wire, as shown on the voltmeter in Fig. 2.2.

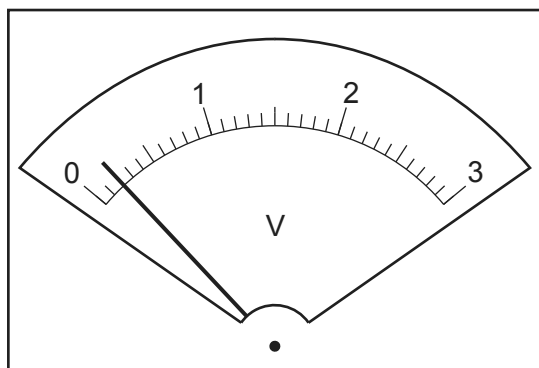


Fig. 2.2

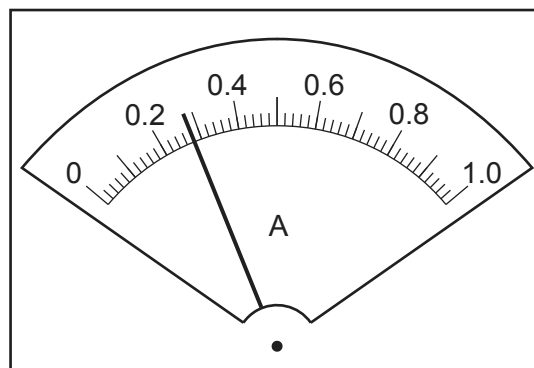


Fig. 2.3

- Record, in the first row of Table 2.1, the current I in the circuit as shown in Fig. 2.3.
- Complete the column headings in Table 2.1.

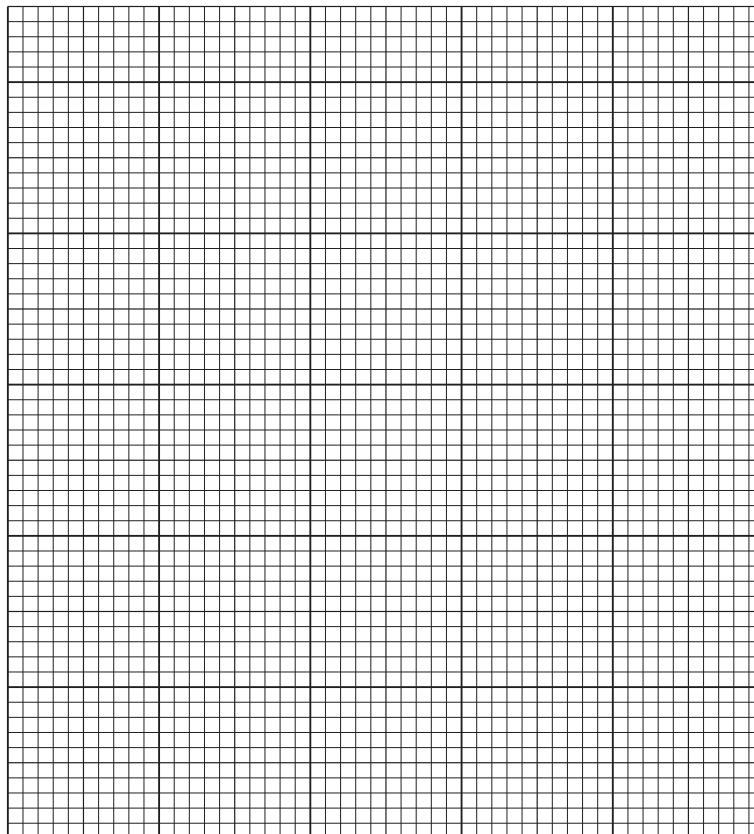
Table 2.1

$l/$	$V/$	$I/$
10.0		
30.0	0.7	0.30
50.0	1.1	0.27
70.0	1.5	0.28
90.0	2.1	0.29

[3]

- (b) The student repeats the procedure using $l = 30.0\text{ cm}$, 50.0 cm , 70.0 cm and 90.0 cm . The readings are shown in Table 2.1.

Plot a graph of V/V (y -axis) against l/cm (x -axis). Start both axes at the origin (0,0).



[4]

- (c) (i) Write a conclusion about the value of the current I in the circuit as the position of the sliding contact C is changed.

.....
 [1]

- (ii) Justify your conclusion by reference to your results.

.....
 [1]

- (d) Using the graph, determine the potential difference V_L when the length $l = 60.0$ cm.

Show clearly on the graph how you obtained your result.

$V_L =$ [2]

[Total: 11]

- 3 A student investigates some thermal properties of sand and water.

Fig. 3.1 shows the apparatus.

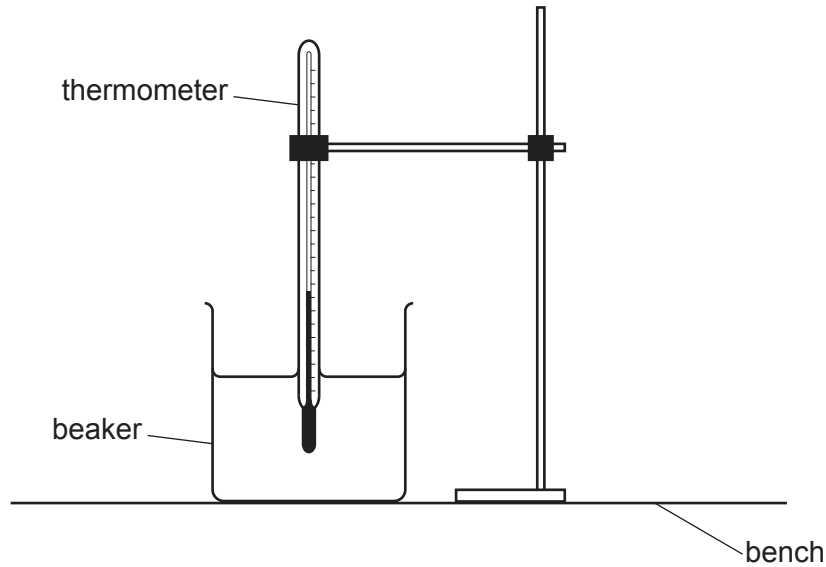


Fig. 3.1

- (a) The thermometer in Fig. 3.2 shows the room temperature θ_S at the beginning of the experiment. Record θ_S .

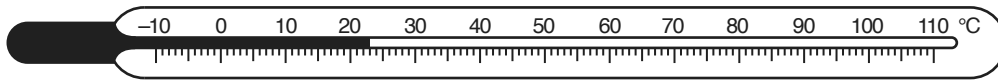


Fig. 3.2

$$\theta_S = \dots\dots\dots\text{ }^{\circ}\text{C} \quad [1]$$

- (b) The student is supplied with hot water at a temperature θ_H . She records the temperature of the hot water.

$$\theta_H = \dots\dots\dots 84^{\circ}\text{C} \dots\dots\dots$$

She pours 100 cm^3 of hot water into a beaker that contains sand. Initially, the sand is at room temperature.

She measures the highest temperature θ_M of the mixture.

$$\theta_M = \dots\dots\dots 70^{\circ}\text{C} \dots\dots\dots$$

- (i) Calculate the rise in temperature θ_R of the sand using the equation $\theta_R = (\theta_M - \theta_S)$.

$$\theta_R = \dots\dots\dots\text{ }^{\circ}\text{C} \quad [1]$$

- (ii) Explain briefly what the student does after pouring the hot water into the sand and before taking the temperature, in order to obtain a reliable value for θ_M .

.....
 [1]

- (iii) Calculate the fall in temperature θ_F of the hot water using the equation $\theta_F = (\theta_H - \theta_M)$.

$$\theta_F = \dots\dots\dots ^\circ\text{C}$$

Calculate the ratio S using the equation $S = \frac{\theta_R}{\theta_F}$. Give your answer to a suitable number of significant figures for this experiment.

$$S = \dots\dots\dots [1]$$

- (c) The student pours 100 cm^3 of the hot water into a clean beaker that contains 100 cm^3 of water at room temperature. She records the highest temperature θ_M of the mixture.

$$\theta_M = \dots\dots\dots 49^\circ\text{C}$$

Calculate the rise in temperature θ_R of the cold water using the equation $\theta_R = (\theta_M - \theta_S)$. Use the value of room temperature θ_S recorded in (a).

$$\theta_R = \dots\dots\dots$$

Calculate the fall in temperature θ_F of the hot water using the equation $\theta_F = (\theta_H - \theta_M)$.

$$\theta_F = \dots\dots\dots$$

Calculate the ratio W using the equation $W = \frac{\theta_R}{\theta_F}$.

$$W = \dots\dots\dots [2]$$

- (d) The student studies the thermal properties of sand and water. She predicts that S should be equal to $6 \times W$.

State whether the results support the prediction. Justify your answer by reference to the readings.

statement

justification

.....

.....

[2]

- (e) Suggest **two** temperatures that it would be sensible to keep constant when carrying out the experiments.

1.

2.

[2]

- (f) The student measures the volume of the dry sand using a measuring cylinder before carrying out the experiment. Tick (✓) the boxes that show the precautions that she should take in order to obtain an accurate reading.

☐

Take the reading at the bottom of the meniscus.

☐

Tap the measuring cylinder to make sure the top of the sand is horizontal.

☐

View the scale of the measuring cylinder at right angles.

[1]

[Total: 11]

- 4 A student investigates the bending of 1 m length strips of different materials. She compares how far they bend when loaded at one end.

Plan an experiment to investigate how the material from which the strips are made affects the bending of the strips when loaded at one end.

The following apparatus is available to the student:

strips of wood, plastic, steel and aluminium, each of length 1 m
a set of slotted masses
a metre rule
a G-clamp (used to hold the strips to the laboratory bench).

Other apparatus normally available in a school laboratory can also be used.

In your plan, you should:

- draw a diagram to show the arrangement of the apparatus
- explain briefly how you would carry out the investigation, including the measurements you would take
- state the key variables to be kept constant
- draw a suitable table, with column headings, to show how you would display your readings (you are **not** required to enter any readings in the table)
- explain how you would use the results to reach a conclusion.

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